

SOLUTIONS - Exercise - Week 2: MATH1061/7861

1. **(5 marks)** Let p, q, r be propositional variables, each either True or False. Consider the following three statements:

$$S_1 = p \vee \sim q$$

$$S_2 = (p \wedge r) \vee q$$

$$S_3 = \sim p \vee (q \wedge r)$$

Task: Using the truth table below find all triples (p, q, r) such that the three statements S_1 , S_2 , and S_3 have the same truth value.

p	q	r	$\sim p$	$p \wedge r$	$q \wedge r$	$p \vee \sim q$	$(p \wedge r) \vee q$	$\sim p \vee (q \wedge r)$	Equal?
T	T	T	F	T	T	T	T	T	Y
T	T	F	F	F	F	T	T	F	N
T	F	T	F	T	F	T	T	F	N
T	F	F	F	F	F	T	F	F	N
F	T	T	T	F	T	F	T	T	N
F	T	F	T	F	F	F	T	T	N
F	F	T	T	F	F	T	F	T	N
F	F	F	T	F	F	T	F	T	N

The statements are only the same for (T, T, T) .

MARKING:

1 mark for a correct concluding statement.

4 marks for the truth table.

No half marks.

2. **(5 marks)** For each of the following, either prove that the statements are logically equivalent using the laws of logical equivalence, or give values which show that they are not equivalent.

Label the laws of logical equivalence used in each step.

(a) $(p \vee \sim q) \vee (\sim p \wedge \sim q) \equiv p$

When p and q are both false, the LHS is true while the RHS is false.

(b) $\sim (p \vee \sim q) \vee (\sim p \wedge \sim q) \equiv \sim p$

$$\begin{aligned}\sim (p \vee \sim q) \vee (\sim p \wedge \sim q) &\equiv (\sim p \wedge q) \vee (\sim p \wedge \sim q) && \text{by De Morgan's law} \\ &\equiv \sim p \wedge (q \vee \sim q) && \text{by the distributive law} \\ &\equiv \sim p \wedge \mathbf{t} && \text{by the negation law} \\ &\equiv \sim p && \text{by the identity law}\end{aligned}$$

MARKING:

2 marks for part (a).

3 marks for part (b). Take off 1 mark if they do not name the laws they use.

No half marks.